

Riemann Sum Progression

A visual progression from a coarse partition to a definite integral, with width, height, and limiting behavior labeled.

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Riemann-sum progression

1. Partition the interval
2. Choose sample points
3. Build rectangle areas
4. Refine the partition
5. Interpret the limiting integral

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Accumulation

$$\Delta x = (b - a)/n$$

$$\sum_{i=1}^n f(x_i^*) \Delta x$$

$$\lim_{n \rightarrow \infty} \sum f(x_i^*) \Delta x = \int_a^b f(x) dx$$